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Overview & Administrative

Group Requirements

Each project will be done by a team of up to 3 students. Every student must register for a group on canvas. This includes both solo and group projects. **Failure to register for a canvas group by the project proposal deadline will result in a zero on your proposal, without exception.**

To register for a group, on canvas go to People -> Project Groups. All groups are labeled as “Term Project XXX”, Do not rename these groups. Prior to submission, ensure all your team members, if any, have joined the same group. Do NOT join other groups without prior communication, Any student that has been found to join another student’s group will be removed from that group and responsible for any resulting point deductions.

When working in a group, please clearly specify in your final report what part/tasks each team member was responsible for.

Submission Instructions

Your project proposal will be submitted through Canvas, Groups will submit one copy for the entire group. Please name your project as `Project_Proposal_Group_XXX.pdf` or `Final_Project_Group_XXX.pdf` where XXX is your project group number registered on canvas. This ensures clarity for your graders. Be sure to include the name of all team members at the beginning of your report.

Project Tips and Guidelines

Your project will be evaluated according to the metrics of evaluating an academic paper. This includes novelty, design, rigorous implementation, and evaluation. A detailed rubric is shown below for both proposals and final projects.

1. Identify a problem related to the course material (Data Analysis, Machine learning related). Your problem doesn't specifically need to utilize material taught in this course, but anything not covered in course content should have appropriate research and citations.

   Examples of past projects:

   a. Predicting used oil price with several factors, using your own models and methods
   b. Testing for differences in daily coffee consumption by an engineering student versus non-engineering student, finding confidence intervals by bootstrap
   c. Building a classification method to predict presidential election outcomes
   d. Finding the most important factor that leads to people becoming friends on Facebook
   e. Finding trends in crime in local neighborhoods

   You can utilize any topics from your current research projects, your dataset doesn't necessarily have to be large, but you should come up with novel ways of using the data. This includes any data found from online sources such as Kaggle. Alternatively, you may also identify a theoretical problem and solve it (Such as proving convergence, consistency, etc.)

2. Collect all your data yourself either by extracting data from the internet or identifying existing dataset sources. Keep in mind that your project should still be novel despite using online datasets, so we highly recommend avoiding commonly used project datasets such as titanic or iris.

3. Analyze your data and fit models using any methods from either the course material or outside research. Carefully design your evaluation method and interpret your results, does your result make sense according to your hypothesis? Was your sample size large enough and representative or biased? Are your findings statistically significant? How could your result be improved? Can it be used in any real-world problems?

Project Academic Integrity

With the constant improvements of GPT tools, we must remind students that your brain is like a muscle, working through projects and problems without offloading that work to GPT helps improve your problem solving and researching capabilities. That said, we expect students will still try to use GPT on their assignments and projects. We remind youthat you must follow all listed policies in the syllabus regarding AI usage, and any violations will result in an OSI investigation. Likewise, we do not allow students to submit the same project to multiple courses without the differences in work being substantial and clearly identified. We utilize Turnitin for checking these potential violations.

Sharing your project

Students often will want to share their project for their portfolio after the semester ends. We do not have any issues with this, but we do require that any public sharing scrubs references to ISYE6740 / CDA from the shared content. This is to limit the potential for future students to easily find your paper online then pass it off as their own, which would potentially trigger a future academic integrity violation to both parties.

Project Proposal

Your project proposal is meant to be a progress check to ensure you're on track. This should include the major sections present in the provided templates. Project proposals must follow these requirements:

- Minimum 2 pages. This must be two reasonable pages, not including things like large half-page graphics/tables, Title pages, or references.
- Font size 11, Single Spaced, Single Column following traditional reporting format

Proposal Grading Rubric

Proposals will be graded utilizing the following rubric. This will be a peer review process, with more details on peer reviews in the peer review section.

2/2 Does this proposal include a problem statement?

1/1 Does this proposal include a proposed methodology?

1/1 Does this proposal include planned evaluation strategies? (no need for results)

1/1 Does this proposal follow formatting requirements (2 Reasonable pages, Single Spaced, etc.)?

Final Project

Your final project will be the culmination of your semesters research into your chosen topic. At a minimum, it should include sections on your Problem Statement, Methodology, and Evaluation/Results. If any significant data cleaning or extraction was needed, you should also include a section on this.

Your final project must be a minimum of 4 reasonable pages, not including things like large half-page graphics/tables, Title pages, or references. You should follow the standard formatting requirements like your proposal: font size 11, Single spaced, single column report format.

Final Project Grading Rubric

Final projects will be graded out of 20 points, utilizing the following rubric through a peer review process. Each section should be graded between 1 and 5

- (5 Points) Creativity and Project Scope
    - Evaluate the project on how unique and innovative it is. Projects that include new ideas or incremental improvements should score higher than replications of existing research. Scope will refer to whether a projects content and complexity accurately reflects the expectations based on group size, i.e. a three-person group should be expected to provide a more detailed and complex project than a 1-person group.
- (5 Points) Formulation
    - Measure of the extent to which the project is grounded in existing theories or models. This will be used to grade the theoretical foundation of the paper, demonstrating the level of understanding regarding the subject matter
- (5 Points) Rigor of Implementation
    - Assesses the technical quality and sophistication of the project's implementation. This will measure how well the project was executed, does the paper demonstrate a level of problem solving in adapting to challenges, and how effectively the project was brought to its final complete state.

- (5 Points) Report Writing Quality
    - This should measure how clearly and coherently the project is communicated. This includes well written arguments and a logical flow through the paper, along with overall presentation such as use of visual aids (charts, graphs, tables), well formatted, and free of grammar errors and typos.

Peer Review Expectations

During this course, Students will be expected to complete Peer Reviews for both their Project Proposal and their Final Project report. This is meant to give students exposure to the types of open-ended feedback present with a true academic paper review and balance the requirements necessary for assigning a grade to a paper.

Your final grade on an assignment will be the average grade of your peer reviews. TA's will also review projects and add additional reviews where necessary to ensure students have appropriate feedback and any outliers are handled.

Peer reviews are assigned automatically after the project deadline for both proposals and final project, and they can be found in canvas by going to the corresponding assignment and clicking "Peer Reviews". Students are expected to enter their peer review as a comment on the assignment and must include both the rubric and quality commentary to receive full credit.

Peer reviews must be submitted by the deadline listed in the syllabus to receive credit, late peer reviews will not be accepted without appropriate accommodations from the dean of student's office.

The following example reviews for both the Proposal and the Final Project are the minimum expectations to receive full credit:

Project Proposal Peer Reviews

Proposal peer reviews will be graded on a 2-point scale. You will receive 1 point for using the proposal review rubric, and the second point for providing quality feedback directly related to the corresponding project as part of your review.

An example of an appropriate proposal Peer Review:

2/2 Project Includes Problem Statement

1/1 Project includes proposed methodology

1/1 Project includes planned evaluation strategies

1/1 Project includes at least two full pages

Really interesting proposal focusing on X topic, exploring Y models looks like an unique approach to take. Have you considered using Y evaluation methodology? Also, What stands out about your specific project to make it unique from other research? Overall, this looks like it will be a great project, be sure to fully explain Z section as it is a bit confusing in this proposal.

Final Project Peer Reviews

Final project peer reviews will be graded on a 4-point scale subject to the following requirements:

- (2 Points) Followed the Final Project Grading rubric, breaking down the given score based on each section of the Rubric.
- (2 Points) Quality comment provided as part of your peer review. This would involve giving at least 1-2 Sentences regarding your fellow students' topic and their specific modeling approach. This can be either included in the above rubric, or as a separate line in your comment after the rubric.

Providing a minimum effort review along the lines of "20/20 good job" will result in only receiving 1 point on your peer review.

An example of a Final Project Peer review that would meet all requirements is as follows:

Total: 18/20
4/5 Creativity and Project Scope: Overall an interesting project, though not enough detail was provided to explain what really makes your project stand out from existing research or analysis.
5/5 Formulation: Great job demonstrating your understanding of how to apply XXX Model!
4/5 Rigor of Implementation: Your overall implementation is good, but I would have liked to see additional follow through on your conclusion. How can this be used, or how can your implementation be repeated?
5/5 Report Writing Quality: Great job on presenting a well formatted paper, the graphs made your results clear to understand!

Overall, a great project! You showed clear understanding of XXX project topic, and the XXX models you explored were an innovative approach. Your final evaluation has led to some interesting conclusions on XXX, which could be further explored.

Difference in Peer Review opinions

In addition, we'd like to address a common occurrence in academic publications and our course – the possibility of contrasting comments from peer reviewers. This variation in feedback is quite common, especially in all real-world publication peer reviews. It is influenced by several factors:

- **Reviewer Expertise:** A reviewer with specialized knowledge in a project's domain may have higher expectations, potentially leading to a more critical assessment.
- **Diverse Understanding:** Conversely, a reviewer less familiar with the project's subject matter might offer feedback that seems less targeted or more general.
- **Student Backgrounds:** In our class, you come from a wide array of academic and professional backgrounds. This diversity naturally leads to a range of perspectives in your reviews.
- **Inherent Subjectivity:** There's always an element of personal bias and variance in any review process.

Rest assured; our peer review system is designed with these considerations. TA's will review both your projects and the peer evaluations. We aim to ensure that the final grades reflect our course standards and are fair and balanced, accommodating the diverse viewpoints and expertise levels of all reviewers.

Thank you for your engagement and thoughtful contributions to the peer review process; Please remember to always use your best judgement when reviewing.